NOT NULL Constraint

What is NOT NULL?

NOT NULL is a SQL constraint that says:

This column **must always have a value**.
Empty (NULL) values are not allowed.

Simple Meaning

If a field is important and should never be missing, use NOT NULL.

Examples:

- employee name
- email
- order date
- product price

What is NULL?

NULL means:

No value / unknown / missing value

It does **not** mean:

- 0
- empty string ''
- false

It means no data stored.

Example Without NOT NULL

```
CREATE TABLE employees (
  id INT,
  name VARCHAR(50),
  salary DECIMAL(10,2)
);
```

Now this is allowed:

```
INSERT INTO employees VALUES (1, NULL, 50000);
```

Name missing.

Bad if name is mandatory.

Example With NOT NULL

```
CREATE TABLE employees (
  id INT NOT NULL,
  name VARCHAR(50) NOT NULL,
  salary DECIMAL(10,2)
);
```

Now this fails:

```
INSERT INTO employees VALUES (1, NULL, 50000);
```

Because name cannot be NULL.

Output Meaning

Database gives error like:

Column 'name' cannot be null

Real-Time Examples

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1. Employees Table

```
CREATE TABLE employees (
  emp_id INT PRIMARY KEY,
  emp_name VARCHAR(100) NOT NULL,
  join_date DATE NOT NULL
);
```

Why?

Every employee must have:

- ID
- Name
- Joining date

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2. Orders Table

```
CREATE TABLE orders (
  order_id INT PRIMARY KEY,
  customer_id INT NOT NULL,
  order_date DATE NOT NULL
);
```

Why?

Order without customer or date is useless.

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3. Banking Table

```
CREATE TABLE accounts (
  acc_no INT PRIMARY KEY,
  customer_name VARCHAR(100) NOT NULL
);
```

No anonymous account records.

Why NOT NULL Is Important

1. Data Completeness

Mandatory fields always filled.

2. Better Reporting

No missing names/emails.

3. Business Logic Safety

Order must belong to customer.

4. Cleaner Analytics

Less dirty data later.

Deep Understanding

When inserting row:

```
INSERT INTO employees(id,name) VALUES (1,NULL);
```

Database checks constraints before saving row.

If name is NOT NULL, insert rejected.

Difference: NULL vs Empty String

name = NULL → no value

name = '' → blank text value

Important distinction.

Some systems allow empty string even with NOT NULL.

Example:

INSERT INTO employees(name) VALUES('');

May succeed because blank string is still a value.

Add NOT NULL to Existing Column

ALTER TABLE employees

MODIFY name VARCHAR(50) NOT NULL;

But first:

All existing NULL values must be fixed.

Check:

SELECT * FROM employees WHERE name IS NULL;

Remove NOT NULL

Allow NULL later:

ALTER TABLE employees

MODIFY name VARCHAR(50) NULL;

NOT NULL vs PRIMARY KEY

Constraint	Meaning
NOT NULL	Value required
PRIMARY KEY	Unique + NOT NULL

So primary key automatically implies not null.

Common Mistake Beginners Make

Using NOT NULL everywhere.

Wrong approach.

Use it only where field is truly mandatory.

Example:

- Middle name may be optional
- Alternate phone optional

Those can allow NULL.

Real Data Engineering Importance

Missing key fields create downstream problems:

- joins fail
- reports wrong
- duplicates harder to detect

So source systems using NOT NULL are healthier.

Interview Answer

Why use NOT NULL?

To ensure mandatory columns always contain data.

Example?

Customer email, employee ID, order date.

Final One-Line Answer

NOT NULL constraint prevents missing values and ensures important columns always have data.

Honest Advice

Understanding NULL behavior deeply is crucial in SQL. Many bugs happen because people misunderstand NULL more than joins.

Primary Key

What is Primary Key?

A **Primary Key** is a column (or group of columns) that **uniquely identifies each row** in a table.

No two rows can have the same primary key value.
Primary key also cannot be NULL.

It is the main identity of each record.

Simple Example

Students Table

student_id	name	city
101	Indra	Bangalore
102	Ravi	Hyderabad
103	Ajay	Chennai

Here:

student_id

is the **Primary Key**.

Because each student has a unique ID.

Why We Need Primary Key

Without primary key:

- duplicate rows possible
- hard to update exact record
- hard to delete exact row
- joins become messy
- no reliable identity

Real Meaning

Think of primary key like:

- Aadhaar number

- Passport number
- Employee ID
- Order ID

Used to uniquely identify one person/record.

SQL Example

```
CREATE TABLE students (
  student_id INT PRIMARY KEY,
  name VARCHAR(50),
  city VARCHAR(50)
);
```

Now:

- student_id must be unique
- student_id cannot be NULL

What Happens If Duplicate Inserted?

```
INSERT INTO students VALUES (101,'Ravi','Hyderabad');
```

If 101 already exists:

Error: Duplicate entry for key PRIMARY

What Happens If NULL Inserted?

```
INSERT INTO students VALUES (NULL,'Ravi','Hyderabad');
```

Fails.

Because primary key cannot be null.

Two Hidden Rules of Primary Key

A primary key automatically means:

1. UNIQUE

No duplicates.

2. NOT NULL

Must always have value.

Deep Internal Importance

Database uses primary key to:

- identify rows quickly
- build indexes
- speed lookups
- connect tables with foreign keys
- maintain data integrity

Example Query

SELECT * FROM students WHERE student_id = 102;

Fast and precise.

Because primary key usually has an index.

Real-Time Examples

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1. Employees Table

emp_id	name
E101	Indra
E102	Ravi

Primary key = emp_id

---

2. Orders Table

order_id	customer	amount
5001	Indra	2500
5002	Ravi	1300

Primary key = order_id

---

3. Banking

acc_no	customer
789456	Indra

Primary key = acc_no

Primary Key vs Normal Column

Feature	Normal Column	Primary Key
Duplicate allowed	Yes	No
NULL allowed	Usually yes	No
Unique identity	No	Yes

Composite Primary Key

Sometimes one column is not enough.

Use two columns together.

Example Attendance Table

student_id	date	status
101	2026-04-23	Present

One student can appear many dates.

So use:

PRIMARY KEY(student_id, date)

Together they uniquely identify row.

Auto Increment Primary Key

Very common.

```
CREATE TABLE students (
  student_id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(50)
);
```

Rows get IDs:

1, 2, 3, 4...

Useful in applications.

Add Primary Key Later

ALTER TABLE students

ADD PRIMARY KEY(student_id);

Drop Primary Key

ALTER TABLE students

DROP PRIMARY KEY;

Important Rules

One Table = One Primary Key

Can contain one or multiple columns, but only one PK definition.

Best Column Choice for Primary Key

Choose column that is:

- unique
- stable (doesn't change often)
- short/simple
- always present

Good:

- employee_id
- order_id

Bad:

- name
- phone number (may change)

Why Not Name as Primary Key?

Because names repeat.

Example:

Two users named Ravi.

So name is poor primary key.

In Data Engineering Why Important?

Primary keys help with:

- deduplication
- joins
- CDC pipelines
- merge/upsert logic
- warehouse modeling

Interview Answer

What is Primary Key?

A primary key uniquely identifies each row in a table and does not allow NULL or duplicate values.

Memory Trick

Primary Key = Main Identity Column

Final One-Line Answer

A primary key is the main column used to uniquely identify every row in a table.

Honest Advice

Understand primary key deeply. It is one of the most important database concepts, more valuable than memorizing many SQL functions.